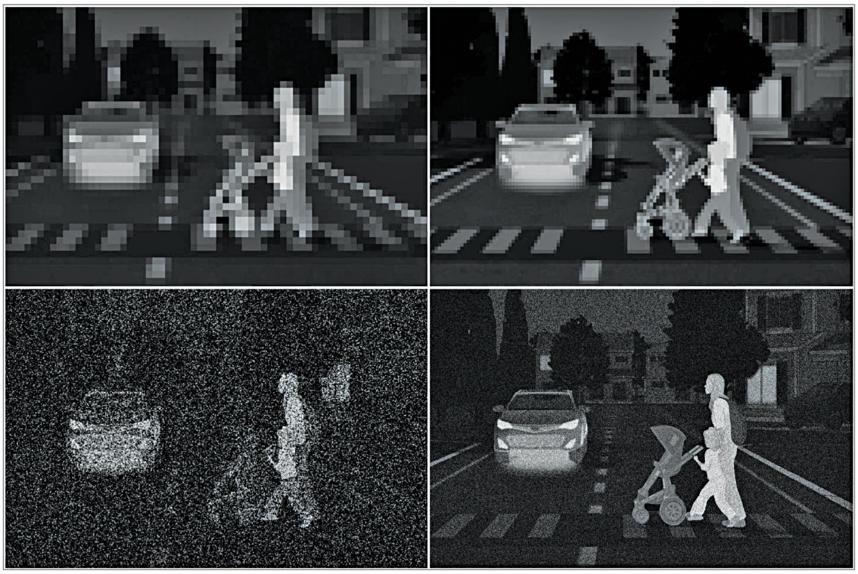
final step is placing this core on a PCB board, adding a lens, and getting it ready to process signals.

The challenge? Packing all this technology into a tiny space without driving up power use and costs.

The end goal of all this tech? To bring pedestrian fatalities down to zero. Big names in the automotive world, like Toyota with their 'Vision Zero' initiative, are on board, aiming for ultra-safe vehicles by 2030.

The key to success? Making this tech affordable and widespread. That is why spreading the word among car makers, suppliers, and even everyday drivers is crucial.



*Thermal cameras for vehicles must be designed to supply necessary data to the CNN. Two important considerations are resolution and thermal sensitivity. On the left are images from cameras with insufficient resolution (top) and thermal sensitivity (bottom) which produce blocky or noisy images. The camera on the right meets both requirements*

## Road safety camera test: A case study

The recent demonstration in Detroit, Michigan, was a significant milestone in showcasing the effectiveness of a state-of-the-art camera system integrated with neural network software. This real-life test, conducted at a test track, was aligned with the expected standards of the National Highway Transportation Safety Authority (NHTSA) in the United States.

***Picture this.*** A car equipped with this high-tech camera system, a test driver, and a pedestrian dummy, all on a test track at night with headlights coming at them.

The big challenge? To see if the camera system could spot the pedestrian and stop the car on its own.

And guess what? It worked! The car detected a child-sized dummy and stopped a safe 6.8 metres away, even in the dark, cold, and windy conditions.

But that's not all. Over in Las Vegas, with its bright lights, another test compared this thermal imaging technology with regular RGB (visible light) imaging. The goal was to see which was better at spotting pedestrians, especially in tricky light conditions.

Turns out, thermal imaging was the clear winner, especially in spotting pedestrians that might be hidden or missed by regular cameras due to backlighting. It even outperformed other tech like lidar and radar in certain situations, like detecting someone near a reflective surface or wearing dark clothing.

So, what does this mean for us? This technology could be a game-changer in making our roads safer, especially in busy urban areas where it's hard to see pedestrians. By integrating these smart cameras into cars, we are looking at a future with fewer pedestrian accidents and safer streets for everyone.

The recent advancements in imaging technology, particularly thermal imaging combined with neural network software, bring us a step closer to a safer world. Through engaging discussions and eye-opening demonstrations in Detroit and Las Vegas, we've seen firsthand the power of this technology to pierce through the veil of night, offering a beacon of hope in low-visibility conditions. These tests are not just technical achievements; they represent a significant leap over traditional imaging and sensor systems in critical safety scenarios. The integration of this advanced technology into vehicles marks the beginning of a transformative era in automotive safety, one that aligns with the goals of global leaders and regulatory bodies. This journey, fuelled by innovation and a collective dedication to saving lives, promises a future where our roads would be safer for pedestrians and drivers alike, especially during the night. It is paving the way to significantly reduce nighttime pedestrian fatalities, making our streets a safer place for everyone. EFY

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